PRAVEEN V 241901082

# Exercise 16
# Footprint a Wireless Network

## AIM

To footprint a wireless network in an authorized lab environment by identifying nearby access points, their SSIDs, BSSIDs, channels, signal strength and encryption types using wireless reconnaissance tools.

## PROCEDURE

1. Connect a wireless network adapter that supports monitor mode to the Kali Linux machine.
2. Open a terminal and list the available wireless interfaces:
   iwconfig
3. Enable monitor mode on the wireless interface:
   airmon-ng start wlan0
4. Verify that the interface is now in monitor mode (wlan0mon):
   iwconfig
5. Scan for nearby wireless networks using airodump-ng:
   airodump-ng wlan0mon
6. Observe the list of access points along with BSSID, channel, encryption type (WEP/WPA/WPA2) and signal strength (PWR).
7. Note the connected client devices (stations) shown at the bottom of the airodump-ng output, along with the BSSID they are associated with.

8. Focus the scan on a specific access point and channel to gather more details:
   airodump-ng -c <channel> --bssid <target-BSSID> -w capture wlan0mon
9. Use Kismet as an alternative tool to passively footprint wireless networks and view additional details such as manufacturer (from MAC OUI) and hidden SSIDs:
   kismet

   Record all discovered access points and their details in the observation table.

**OBSERVATION TABLE**

| S.No | SSID | BSSID | Channel | Encryption | Signal Strength (PWR) | Connected Clients |
|---|---|---|---|---|---|---|
| 1 | Lab_WiFi | AA:BB:CC:11:22:33 | 6 | WPA2 | -40 | 2 |
| 2 | Lab_Guest | AA:BB:CC:44:55:66 | 11 | WPA2 | -55 | 0 |
| 3 | TestNet_Open | AA:BB:CC:77:88:99 | 1 | Open | -65 | 1 |
| 4 | Hidden_AP | AA:BB:CC:AA:BB:CC | 3 | WPA2 | -70 | 0 |

**OUTPUT**

**airodump-ng scan output:**

BSSID PWR CH ENC ESSID

AA:BB:CC:11:22:33 -40 6 WPA2 Lab_WiFi

AA:BB:CC:44:55:66 -55 11 WPA2 Lab_Guest

AA:BB:CC:77:88:99 -65 1 OPN TestNet_Open

AA:BB:CC:AA:BB:CC -70 3 WPA2 <length: 0>

BSSID STATION PWR Rate Lost Frames Probe

AA:BB:CC:11:22:33 11:22:33:44:55:66 -42 1e- 1e 0 15

**RESULT**

Thus, footprinting of a wireless network was successfully performed in the authorized lab environment. The access points, their SSIDs, BSSIDs, channels, encryption types and connected clients were identified using airodump-ng and Kismet.